

Problem Domain, Dataset & Methodology

Spatiotemporal Forecasting under Domain Shift using RAConv

COMP 4360 - Machine Learning | University of Manitoba

This document details the problem domain, synthetic dataset generation pipeline,
and the methodology & adaptation strategy used in this project.

Part 1: Problem Domain & Dataset

1.1 Motivation and Domain Shift

The original RAConv architecture (Wang & Wong, IEEE ICC 2022) was designed for cellular network traffic prediction, where base stations generate spatiotemporal load patterns with daily/weekly cycles. The model learns: (1) local short-term spatiotemporal patterns from neighbor interactions, and (2) long-term dependencies from recurring temporal cycles.

This project transfers the same modeling assumptions to epidemiology, a fundamentally different domain. In the epidemiological setting:

- Each spatial cell represents a city rather than a base station
- Temporal dynamics reflect disease progression (S->E->I->R) rather than network load cycles
- Spatial coupling reflects human mobility and disease spread rather than handoff interference
- Regime changes correspond to policy interventions (lockdowns, reopenings) rather than traffic events

This constitutes a clear domain shift: while the data structures are analogous (both are spatiotemporal grids evolving over time), the underlying generative processes are entirely different. The project evaluates whether RAConv's inductive biases, residual 3D convolutions for local spatiotemporal features and attention-based ConvLSTM for long-range dependencies, can generalize across this shift.

1.2 Geographic Foundation: 256 Texas Cities

The spatial domain consists of the 256 most populous cities in Texas, selected from the SimpleMaps `uscities.csv` dataset. Texas was chosen for its large population (~31.9 million across these cities), geographic diversity (spanning latitudes 25.8-36.5 N and longitudes -106.6 to -93.5 W), and well-documented COVID-19 policy timeline.

Data Collection (Task 1)

Cities are sorted by population descending and the top 256 are retained. For each city, the pipeline stores: city name, population, latitude, and longitude. A 256x256 pairwise distance matrix is computed using haversine distances (multiplied by Earth's radius 6,371 km), yielding distances in kilometers. The distance matrix is validated for symmetry and triangle inequality.

Data Validation (Task 2)

The preprocessing validates: (1) total population is in the expected 15-34 million range, (2) coordinate bounds fall within Texas boundaries, (3) no duplicate city names exist, (4) the distance matrix is symmetric to within 1e-6 tolerance, and (5) the triangle inequality holds for a sample of 50 cities.

1.3 Gravity-Based Mobility Matrix (Task 3)

Inter-city movement is modeled using a gravity model, a standard approach in spatial epidemiology (Balcan et al., PNAS 2009). The raw flow between cities i and j is:

$$\text{theta}_{ij} = N_j / D_{ij}^{\alpha} \quad (\text{for } i \neq j)$$

where N_j is the destination city population, D_{ij} is the haversine distance in km, and $\alpha = 2.0$ is the distance decay exponent. Each row is then normalized so that the total daily outflow from city i equals 2% of its population (the baseline

per-capita inter-city movement rate). This produces a 256x256 asymmetric flow matrix where larger cities attract more inflow and distant cities exchange fewer travelers.

Validation confirms: all cities have 1-3% daily outflow rates, large cities (Houston, San Antonio, Dallas, Austin, Fort Worth) dominate inflow, and flow magnitude decays with distance at the expected rate.

1.4 Coupled SEIR Metapopulation ODE System (Task 4)

The core epidemiological simulation implements a coupled SEIR (Susceptible-Exposed-Infectious-Recovered) metapopulation model, where 256 cities exchange infected and susceptible individuals through the mobility matrix. The system of ordinary differential equations for each city i is:

$$\begin{aligned} dS_i/dt &= -\beta(t) * S_i * I_i/N_i + \text{mobility_S_net} \\ dE_i/dt &= \beta(t) * S_i * I_i/N_i - \sigma * E_i + \text{mobility_E_net} \\ dI_i/dt &= \sigma * E_i - \gamma * I_i + \text{mobility_I_net} \\ dR_i/dt &= \gamma * I_i + \text{mobility_R_net} \end{aligned}$$

This gives a total state dimension of 256 cities x 4 compartments = 1,024 coupled ODEs, integrated using SciPy's RK45 solver with $\text{rtol}=1\text{e-}6$ and $\text{atol}=1\text{e-}9$ over 300 simulated days.

Fixed Epidemiological Parameters

Parameter	Value	Meaning	Source
sigma	0.192 /day	$1/\sigma = 5.2\text{d incub.}$	Linton et al. 2020
gamma	0.100 /day	$1/\gamma = 10\text{d infect.}$	He et al. 2020
alpha	1.10	Nonlinear transmiss.	Allee suppression
Report. frac.	10%	$I_{\text{rep}} = I \times 0.10$	TX CDC seroprev.

Time-Varying Transmission Rate $\beta(t)$

The transmission rate $\beta(t)$ is piecewise-constant, calibrated to Texas COVID-19 policy events anchored to Day 0 = March 6, 2020 (first confirmed TX community case). The schedule was designed to produce two visible epidemic waves with realistic attack rates.

Days	Date Range	beta	R0	Phase
0-5	Mar 6-11	0.30	3.0	Uncontrolled spread
5-25	Mar 11-31	0.26	2.6	Voluntary distancing + school closures
25-56	Mar 31-May 1	0.12	1.2	Stay-at-home order (partial)
56-89	May 1-Jun 3	0.16	1.6	Phase 1-2 reopening
89-111	Jun 3-25	0.20	2.0	Phase 3 (restaurants 75%)
111-118	Jun 25-Jul 2	0.17	1.7	Bars re-closed, rollback
118-165	Jul 2-Aug 18	0.09	0.9	Mask mandate ($R(t) < 1$)
165-195	Aug 18-Sep 17	0.10	1.0	Post-Wave-1 trough
195-215	Sep 17-Oct 8	0.12	1.2	75% capacity: Wave 2 trigger
215-265	Oct 8-Nov 27	0.13	1.3	Wave 2 growth (fatigue)
265-283	Nov 27-Dec 14	0.15	1.5	Thanksgiving amplification
283-300	Dec 14-Jan 1	0.14	1.4	Vaccines begin; W2 peak approach

Crucially, Wave 2 uses LOWER beta than Wave 1 but produces a LARGER epidemic peak. This realistic behavior emerges because: (1) all 256 cities are seeded by Wave 2 (vs. Houston only at Day 0), (2) the larger initial infected pool accelerates absolute growth even at lower R_0 , and (3) behavioral fatigue reduces mobility restrictions compared to Wave 1.

Time-Varying Mobility Scale

The mobility matrix is scaled by a time-varying factor calibrated to Google Community Mobility Reports for Texas. The pre-pandemic baseline is 3%/day inter-city movement. During the stay-at-home order (Days 25-56), mobility drops to 1.5%/day (-50%). During Wave 2 (Days 215-265), mobility only drops to 2.5%/day (-17%), roughly half the Wave 1 reduction, reflecting documented behavioral fatigue (IHME, Nature Medicine 2021).

Nonlinear Feedback Mechanisms

The ODE system includes three nonlinear feedback mechanisms beyond standard SEIR: (1) State-dependent behavioral feedback on transmission: higher statewide prevalence lowers effective beta via $\text{beta_eff} = \text{beta} / (1 + K * \text{prevalence}^P)$ with $K=20$. (2) Nonlinear (Allee-like) transmission: infection is suppressed at very low local prevalence but recovers standard dynamics above a reference threshold (0.8% local prevalence). (3) Local mobility suppression: cities with higher infection prevalence reduce outgoing movement via an exponential dampening factor $\exp(-6 * \text{local_prevalence})$.

Initial Conditions & Seeding

Houston (population ~6.05 million, the largest TX city) is seeded with 1,000 initial infected cases at Day 0. All other cities start fully susceptible. A 1,000-person undetected seed is realistic for a COVID-like outbreak (given under-detection at the start of a pandemic).

1.5 Spatial Grid Mapping & Model Input Preprocessing

Geographic Snake-Sort Grid Assignment

The 256 cities are mapped to a 16x16 spatial grid using a latitude-sorted snake pattern that preserves geographic neighborhood relationships. The algorithm: (1) Sort all 256 cities by latitude, (2) partition into 16 rows of 16 cities each, (3) within each row, sort by longitude alternating direction (left-to-right for even rows, right-to-left for odd rows). This snake pattern ensures that adjacent grid cells correspond to geographically proximate cities, which is critical for the convolutional layers to capture meaningful spatial correlations.

Feature Extraction: Infectious (I) Compartment Only

From the full SEIR output (S, E, I, R for 256 cities over 301 time steps), only the Infectious compartment (I) is extracted as the model's learning target. This produces a 3D tensor of shape (301, 16, 16) representing the spatial distribution of active infections over time. The I compartment is the most informative for forecasting because it directly drives transmission dynamics and is the quantity most relevant to public health response.

Train/Validation/Test Split

The 301-day time series is split chronologically (no shuffling, to prevent temporal leakage): Train: first 180 days (60%), Validation: days 180-240 (20%), Test: days 240-300 (20%). The train split covers Wave 1 growth through initial NPI period. The test split covers the critical Wave 2 dynamics, providing a natural out-of-distribution evaluation.

Global Min-Max Normalization

A global (scalar) min-max normalization is fit on the training segment only, then applied to all splits. This ensures: (1) no data leakage from validation/test into normalization statistics, (2) values in [0,1] range for stable gradient flow, and (3) an exact inverse transform is available for computing real-scale error metrics at evaluation time.

Sliding Window Sampling

Supervised training pairs are created using a sliding window with stride 1. For lookback window P and forecast horizon Q=7:

```
X: (N, P, 16, 16)  -- past P days of normalized I
Y: (N, Q, 16, 16)  -- next 7 days of normalized I
```

Multiple lookback windows P are tested: {4, 6, 8, 10, 14} days. The preprocessing generates separate .npz files for each P value, all sharing Q=7 as the prediction horizon.

Dataset Sizes (Example: P=14, Q=7)

Split	Samples	X Shape	Y Shape
Train	160	(160,14,16,16)	(160,7,16,16)
Validation	40	(40,14,16,16)	(40,7,16,16)
Test	41	(41,14,16,16)	(41,7,16,16)

Part 2: Methodology & Adaptation

2.1 Original RAConv Architecture

The RAConv architecture (Wang & Wong, IEEE ICC 2022) was developed for cellular traffic prediction. The original pipeline: (1) 4,096 base stations clustered into 16 spatial groups, (2) each group trains an independent RAConv model, (3) the model predicts future traffic load frames from a history of spatial traffic maps.

The architecture has three core components working in sequence: a 3D convolutional residual front-end that extracts hierarchical spatiotemporal features, an attention-enhanced ConvLSTM that captures long-range temporal dependencies with spatial attention, and an autoregressive decoder that generates multi-step future predictions.

2.2 Adaptation Strategy: Traffic to Epidemiology

The key adaptation decisions made for transferring RAConv to the epidemiological domain:

Single-Cluster Design

In the original paper, 4,096 base stations are clustered into 16 groups, each modeled separately. In our adaptation, all 256 cities are treated as ONE spatial cluster mapped to a single 16x16 grid. A single shared RAConv model is trained over the entire state. This reduces computational cost while preserving spatial structure. The setup remains comparable since the original paper reports results from a single local model.

Input Representation

Cellular traffic: (B, 1, T, H, W) where each frame is a spatial load map with values in [0,1] after normalization. Epidemic adaptation: (B, 1, P, 16, 16) where each frame is the spatial distribution of active infections I, min-max normalized. The video-like tensor format is identical; only the semantic meaning of the spatial signal changes.

Forecast Horizon

The original model predicts Q future traffic frames. Our adaptation fixes Q=7 (7-day forecast), which is operationally relevant for epidemic response planning (aligns with weekly reporting cycles in public health). Multiple lookback windows P in {4, 6, 8, 10, 14} are tested to determine optimal historical context length.

2.3 RAConv Architecture (As Implemented)

Stage 1: Initial 3D Convolution

Input tensor (B, 1, P, 16, 16) is passed through a Conv3d(1->64, kernel=3, padding=1) followed by BatchNorm3d and ReLU. This lifts the single-channel input to a 64-channel feature volume while preserving spatial and temporal dimensions. Output: (B, 64, P, 16, 16).

Stage 2: Three ResBlock3D Modules (ResConv3D)

Three residual blocks progressively widen the channel dimension: ResBlock3D(64->64), ResBlock3D(64->96), ResBlock3D(96->128). Each ResBlock3D contains 5 Conv3D layers organized as two sub-blocks with skip connections:

- Conv3D -> BN -> [shortcut 1] -> Conv3D -> BN -> ReLU -> Conv3D -> BN -> Add(shortcut 1) -> ReLU
- [shortcut 2] -> Conv3D -> BN -> ReLU -> Conv3D -> BN -> Add(shortcut 2) -> ReLU -> Output

The residual connections stabilize training for deeper architectures and preserve spatial detail. All convolutions use kernel_size=3, stride=1, padding=1, maintaining (B, C, P, 16, 16) shape throughout. After all three blocks, the feature volume has shape (B, 128, P, 16, 16).

Stage 3: Attention ConvLSTM (AConvLSTM)

The 128-channel spatiotemporal feature volume is permuted to (B, P, 128, 16, 16) and fed frame-by-frame into a 2-layer stacked AConvLSTM with hidden_channels=[256, 256] and kernel_size=[3, 3].

The attention mechanism replaces standard sigmoid gates on the input and output gates with learned spatial attention maps. For the input gate: the pre-activation ($W_{xi} * X + W_{hi} * H_{prev}$) is passed through tanh, then a 1x1 convolution (W_{i_att}), followed by a numerically stable exp/max normalization that produces attention weights in (0, 1]. The same mechanism is applied to the output gate. The forget gate retains standard sigmoid gating with peephole connections.

This attention mechanism allows the model to dynamically focus on spatially important regions, which is critical in epidemiology where infection hotspots are highly localized and shift over time.

Stage 4: Autoregressive Multi-Step Prediction

After encoding the full P-step history, the last hidden states from both LSTM layers are used as the initial state for autoregressive decoding. The process:

- Take the 128-channel feature map from the last time step as first_input
- For each of Q=7 future steps: pass input through both LSTM layers, project hidden state to 1-channel prediction via Conv2d(256->1, kernel=1)
- Feedback bridge: the 1-channel prediction is projected back to 128 channels via Conv2d(1->128, kernel=1) and fed as input for the next step

This produces the final output tensor of shape (B, 7, 1, 16, 16), representing the predicted infection maps for the next 7 days.

2.4 Ablation Model: AConvLSTM-Only

To quantify the contribution of the ResConv3D front-end, an ablation model removes all residual blocks and feeds raw input directly into the AConvLSTM layers. Architecture: Input (B, P, 1, 16, 16) -> AConvLSTMLayers(input_channels=1, hidden=[256,256], kernel=[3,3]) -> autoregressive decode for Q steps.

Property	RAConv (Full)	AConvLSTM (Ablation)
3D Conv Init	Yes (1->64)	No
ResBlock3D	3 blocks (64->128)	None
AConvLSTM Input Ch.	128	1
Hidden Channels	[256, 256]	[256, 256]
Attention Gates	Yes	Yes
Parameters	~12.35M	~7.35M
Feedback Bridge	128-ch feature	1-ch raw pred

2.5 Training Configuration

Both models share the same training protocol for fair comparison:

Hyperparameter	RAConv	AConvLSTM
Loss Function	MSE	MSE
Optimizer	Adam	Adam
Learning Rate	1e-3	1e-3
Weight Decay	1e-4	1e-4
LR Scheduler	ReduceLROnPlateau	ReduceLROnPlateau
factor / patience	0.5 / 5 epochs	0.5 / 5 epochs
Batch Size	16	16
Max Epochs	50	50
Gradient Clipping	max_norm=1.0	max_norm=1.0
Dropout	0.0	0.2
Checkpoint	Best val MSE	Best val MSE

2.6 Evaluation Methodology

Models are evaluated on the held-out test set (Days 240-300), which covers the critical Wave 2 dynamics that the model has never seen during training.

Metrics

- MSE (Mean Squared Error) - primary training objective, both normalized and real-scale
- MAE (Mean Absolute Error) - more interpretable error magnitude
- RMSE (Root Mean Squared Error) - in normalized space and real scale (infected persons)
- Per-step MSE/MAE - error breakdown for each of the 7 forecast days
- Spatial MSE map - 16x16 heatmap showing where errors concentrate geographically

Inverse Transform for Real-Scale Metrics

Since models operate on min-max normalized data, predictions are inverse-transformed using the training-fit min/max scalars before computing real-scale metrics. This allows reporting error in units of infected persons (e.g., RMSE = 3,229 infected), which is interpretable for epidemiological applications.

2.7 Results Summary

The full RAConv model consistently outperforms the AConvLSTM-only ablation across all lookback windows, demonstrating that the ResConv3D front-end provides significant value for spatiotemporal epidemic forecasting.

P	RAConv MSE	AConv MSE	RAConv RMSE	AConv RMSE	RAConv RMSE*	AConv RMSE*
4	0.00268	0.00773	0.0518	0.0879	4,558	7,746
6	0.00185	0.00770	0.0430	0.0877	3,775	7,724
8	0.00156	0.00495	0.0395	0.0704	3,482	6,136
14	0.00136	0.00281	0.0368	0.0530	3,229	4,453

* RMSE in real scale (infected persons). Lower is better.

Key Findings

- RAConv achieves 27-51% lower real-scale RMSE than the AConvLSTM-only baseline across all P values.
- Longer lookback (P=14) is best for both models, with RAConv P14 achieving RMSE=3,229 infected.
- The ResConv3D front-end contributes ~5M additional parameters but delivers substantial accuracy gains, validating that hierarchical spatiotemporal feature extraction benefits epidemic forecasting.
- Per-step error analysis shows RAConv maintains relatively stable error across all 7 forecast steps, while AConvLSTM error grows more rapidly with forecast horizon.
- The domain shift from cellular traffic to epidemiology is successful: RAConv's inductive biases (local 3D convolutions + attention LSTM) transfer meaningfully to disease spread dynamics.

References

- [1] Wang, Z. & Wong, V.W. (2022). Cellular Traffic Prediction Using Deep Convolutional Neural Network with Attention Mechanism. IEEE ICC.
- [2] Shi, X. et al. (2015). Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting. NeurIPS.
- [3] Keeling, M.J. & Rohani, P. (2008). Modeling Infectious Diseases in Humans and Animals. Princeton University Press.
- [4] Balcan, D. et al. (2009). Multiscale mobility networks and the spatial spreading of infectious diseases. PNAS.
- [5] Yu, D. et al. (2021). Assessing effects of reopening policies on COVID-19 in Texas. Infectious Disease Modelling 6, 461-473.
- [6] Oden, J.T. et al. (2020). Impact of Social Distancing on COVID-19 Healthcare Demand. Emerg Infect Dis.
- [7] Google COVID-19 Community Mobility Reports (Texas), Feb 2020 - Jan 2021.
- [8] IHME (2021). Modeling COVID-19 scenarios for the United States. Nature Medicine 27, 51-63.
- [9] Linton, N.M. et al. (2020). Incubation Period and Other Epidemiological Characteristics. J Clin Med.
- [10] He, X. et al. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. Nature Medicine.